National Tsing Hua University
Department of Electrical Engineering
EE4292 IC Design Laboratory (積體電路設計實驗)
Fall 2025

Term Project (30%)

Assigned on Nov 13, 2025
Due by Dec 26, 2025

Objectives:

Based on the knowledge acquired in this course, you need to 1) implement a target algorithm into RTL codes and 2) do a basic APR flow to generate its layout and evaluate chip performance. The topic can be any digital implementation (see details below) that meets a complexity threshold, subject to both subjective evaluation by the instructor and a minimum area requirement of 150K μm^2. The proposal and final report should include discussions on four key aspects: functionality, specification, implementation, and verification.

Project Examples:

1. **Enhancement to the CPU in labs** (default difficulty: low-medium)
   Try to enhance the CPU in non-trivial ways. For example, you can handle data hazard completely by hardware, instead of by the workaround NOP insertion. You can also add multiplication and make it five-stage pipelined. Other kinds of complex instructions can also be added. For this topic, you have to give a solid test plan to verify your design.

2. **Advanced CNN Accelerator** (default difficulty: medium-high)
   Try to implement a hardware accelerator supporting AI models larger than HW5 or new data formats, e.g. block floating-point formats such as OCP Microscaling (MX) or NVFP4. You may discuss the trade-off between hardware complexity and model quality.

3. **Floating-point ALU** (default difficulty: low-high)
   Try to implement floating-point arithmetic operations. For example, you could support on-the-fly options for **FP64/FP32/FP16** in the IEEE 754-1985 format, or you could consider the new posit format (https://en.wikipedia.org/wiki/Unum_(number_format)#Unum_III) for advanced study. You could further support (**recommended**) some usual analytic functions based on this ALU, such as sin, cos, log, *etc*.

4. **Matrix Multiply Unit** (default difficulty: low-medium)

Try to implement an efficient hardware accelerator for matrix multiplication which is a key operation in deep learning. You may refer to the matrix multiply unit in Google TPU. Discussion on efficient implementation to support self-attention, the core algorithm in state-of-the-art deep learning models, is encouraged and **recommended**.

5. **2048-point Fast Four Transform** (default difficulty: **low**)
   FFT is **not encouraged** in this year, unless you build a larger system based on it.

6. **JPEG** (default difficulty: **low**)
   JPEG is **not encouraged** in this year, unless you build a larger system based on it.

Teaming:

Each team should consist of **two to three** members, and the grading will depend on the contribution of each member (if listed).

Project Schedule:

Nov 13, 2025:	Project announcement
Nov 27, 2025:	Project proposal (topic and team member identified; check with TA 張智鈞)
Nov 29, 2025:	Project difficulty assigned
Dec 11, 2025:	Interim presentation; front-end design: software (if any), RTL and testbench
Dec 26, 2025:	Final presentation; submission of your design and project report (no late submission is allowed)

Grading Rule (100%):

1. Completeness (25%)
   - Basic flow: RTL code, testbench, RTL simulation, synthesis, APR, post-layout simulation
2. Difficulty (25%)
   - High: 20-25%; Medium: 10-20%; Low: 5-10%
   - **Note**: Floorplan utilization, clock speed, and core area all affect this metric. For example, you may get assigned as Low for having low utilization (e.g. $\leq$50%), low speed constraint (e.g. 100ns), or small core area (e.g. $\leq$150Kum^2).
3. Presentation (25%)
   - Interim: 10%
   - Final: 15% (10% by instructor & TA, 5% by students' cross-evaluation)
4. Documentation (25%)
   - Proposal (3%)
   - Final report (22%)

Deliverable (per team basis):

1. Project proposal briefly describing the following: (see the template for reference)

- Team members
- Your topic*

2. Interim presentation slides (in three minutes) including:
   - Functionality and specification of your project
   - Current status of your project on testbench and RTL design
   - **Note**: Justify your specification for a target application.
3. Final presentation slides (in four minutes)
   - **Note 1**: Focus on novelty and contributions. Leave details to the final report.
   - **Note 2**: Do not just dump figures and data. Explain them with reasoning.
4. Project report summarizing: (in 5 pages, either single- or double-column)
   - Your result*
   - Effort of each member
   - **Note**: Clearly specify your novelty and indicate which portions do not belong to your own work for this term project.
5. Verilog code and EDA tool scripts

*Note – Problem-based designs are expected, and four aspects should be covered:

Aspect	Proposal	Final Report
Functionality	Initial ideas of your design (what it should be)	Implemented ideas (what it really is)
Specification (Area/Speed/Power)	Target performance (how well it should be)	Chip performance (how well it really is)
Implementation	Initial plan (what you will do)	Achieved items (what you really have done)
Verification	Test plan (how to verify functionality and specification)	Verification results